

Nash equilibrium

1. A Nash equilibrium is a strategy profile in which ...
 - (a) no player ever randomises
 - (b) every player's strategy is a best response to the others' strategies
 - (c) every player plays a strictly dominant strategy
 - (d) the sum of the players' payoffs is maximised
2. A strategy s^* for player i is a best response to the others if ...
 - (a) it gives every player the same payoff
 - (b) it is the strategy with the largest support
 - (c) it gives player i at least as high a payoff as any other strategy, given the others
 - (d) it strictly dominates every other strategy
3. In Rock-Paper-Scissors, the unique Nash equilibrium is for each player to ...
 - (a) play each action with probability $1/3$
 - (b) copy the opponent's last move
 - (c) always play Rock
 - (d) there is no Nash equilibrium
4. The support of a mixed strategy is ...
 - (a) the set of actions played with positive probability
 - (b) the action with the highest payoff
 - (c) the opponent's best response
 - (d) the set of all of the player's actions
5. An action is strictly dominated if ...
 - (a) it is never a best response in a pure equilibrium
 - (b) the opponent also has a dominated action
 - (c) it gives a lower payoff than some other action against at least one opponent strategy
 - (d) some strategy gives a strictly higher payoff against every strategy of the opponents
6. By the best response condition, a mixed strategy is a best response if and only if ...
 - (a) the opponent is also indifferent
 - (b) all actions are played with equal probability
 - (c) it places all weight on a single action
 - (d) every action in its support yields the maximal expected payoff

7. A strictly dominant action for a player is one that ...
- (a) is a best response to at least one opponent strategy
 - (b) maximises the combined payoff of both players
 - (c) gives a strictly higher payoff than every other action, against every opponent profile
 - (d) is played in some Nash equilibrium
8. A pure strategy is a special case of a mixed strategy in which ...
- (a) all the probability is placed on a single action
 - (b) the player has only one action
 - (c) the opponent is indifferent
 - (d) every action is equally likely
9. Nash's theorem guarantees that every finite game has ...
- (a) no Nash equilibrium unless there is a dominant strategy
 - (b) a Nash equilibrium only if it is zero sum
 - (c) a unique Nash equilibrium
 - (d) at least one Nash equilibrium, possibly in mixed strategies
10. A coordination game (such as choosing the same side of the road) typically has ...
- (a) infinitely many pure equilibria
 - (b) no Nash equilibrium
 - (c) a single dominant-strategy equilibrium
 - (d) two pure equilibria and one mixed equilibrium
11. To find the mixed equilibrium of a 2x2 game you ...
- (a) eliminate the dominant strategy
 - (b) maximise each player's own payoff
 - (c) set both players to play uniformly
 - (d) choose each player's mix so the other player is indifferent between their actions
12. A two-player game is non-degenerate if ...
- (a) the payoff matrices sum to zero
 - (b) no strategy of support size k has more than k best response actions
 - (c) every player has a strictly dominant strategy
 - (d) it has a unique Nash equilibrium
13. An action that is weakly (but not strictly) dominated ...

- (a) is never a best response
 - (b) can still be a best response against some opponent strategy
 - (c) is always played in equilibrium
 - (d) must be strictly dominated too
14. The support enumeration algorithm finds Nash equilibria by ...
- (a) removing all dominated actions
 - (b) checking each possible pair of supports for the indifference and best response conditions
 - (c) simulating the game many times
 - (d) computing the value of the game
15. If a game has a strictly dominant strategy for each player, then ...
- (a) there is no Nash equilibrium
 - (b) the equilibrium must be mixed
 - (c) the profile of dominant strategies is the unique Nash equilibrium
 - (d) there are many Nash equilibria
16. The existence of a Nash equilibrium in every finite game is proved using ...
- (a) the central limit theorem
 - (b) the fundamental theorem of algebra
 - (c) Kakutani's fixed-point theorem applied to the best response correspondence
 - (d) linear programming duality
17. The map $\beta_i(s_{-i}) = \arg \max_{\sigma_i} u_i(\sigma_i, s_{-i})$ is called the ...
- (a) payoff matrix
 - (b) best response correspondence
 - (c) replicator equation
 - (d) support
18. A mixed strategy for a player is ...
- (a) a probability distribution over that player's actions
 - (b) a payoff value
 - (c) an ordering of the opponents
 - (d) a single chosen action
19. In the support enumeration algorithm, for a candidate pair of supports one solves equations making each player ...

- (a) prefer their first action
 - (b) minimise the opponent's payoff
 - (c) indifferent between the actions in their own support
 - (d) play a pure strategy
20. Nash's theorem guarantees at least one equilibrium but ...
- (a) never a mixed one
 - (b) always exactly one
 - (c) only in zero-sum games
 - (d) not necessarily a unique one